APTITUDE [PROFIT & LOSS]

Q) Given:

Buys for ₹400 [CP]

sells for ₹480 [SP]

Profit ⇒ 480 − 400 [SP − CP].

⇒ 80.

$$\text{Profit \%} \Rightarrow \frac{\text{Profit}}{\text{CP}} \times 100$$

$$\Rightarrow \frac{\cancel{80}^{20}}{\cancel{400}} \times \cancel{100}$$

⇒ 20%

Profit percentage = 20%

Q) Given:

CP ⇒ ₹1500

SP ⇒ ₹1200

Loss ⇒ CP − SP

⇒ 1500 − 1200

⇒ 300

$$\text{Loss \%} \Rightarrow \frac{\text{Loss}}{\text{CP}} \times 100$$

$$\Rightarrow \frac{\cancel{300}^{20}}{\cancel{1500}_{\cancel{5}\,1}} \times \cancel{100}$$

⇒ 20%

Loss % = 20%

Dr) ~~Cost~~ CP ⇒ + 20 %

Discount = 10 %

PP = ?

Assume CP = 100

* Marked 20 % above CP.

∴ 20 % of 100 = 20

So, MP = 100 + 20 = 120

* 10 % discount on MP.

10 % of 120 = $\frac{10}{100} \times 120$

= 0.10 × 120

= 12

So, SP = 120 − 12

= 108.

* Profit = SP − CP

= 108 − 100

= 8.

* PP ⇒ $\frac{\text{Profit}}{\text{CP}} \times 100$

= $\frac{8}{100} \times 100$

= 8 %

PP = 8 %

Q) CP of 5 pens = ₹100

SP of each pen = ?

to make 25% profit

for 1 pen $CP = \frac{100}{5}$

$= ₹20$

To make 25%

$SP = CP \times \left(1 + \frac{25}{100}\right)$

$= 20 \times 1.25$

$= ₹25$.

SP for each pen = ₹25.

Q) SP = ₹900

Loss % = 10.

CP = ?

SP = 90% of CP

sy with 10% loss

900 = 90% of CP.

$\cancel{CP = \left[\frac{SP}{1 - \frac{L}{100}}\right]}$

$900 = \frac{90}{100} \times CP.$

$CP = \cancel{900}^{10} \times \frac{100}{\cancel{90}}$

$\cancel{= \frac{900}{1 - \frac{10}{100}}} \Rightarrow$

$= 1000$

CP = ₹1000